Tectonic hazards knowledge organiser

Tuesday 23 February 2027 | P2 | 40 minutes | Year A

Humanities | LAUNCH | Spring 2 Week 1 | HUM-LAUNCH-S2-W01

White paper version. Read quietly, point, write, type, speak privately or direct a scribe who reads your words back. You may pass. No public answer or reading aloud is required.

Learning outcome: Explain causes of tectonic hazards.

Earth’s stiff outer layer is split into plates. They move slowly over mantle rock. Most mantle rock is solid. A plate boundary is where plates meet. Many hazards occur near these edges.

Movement	Earthquake link	Volcano link
Apart or divergent	Cracks and slips can cause shallow earthquakes.	Hot rock rises. Lower pressure allows some rock to melt. Magma may erupt.
Together with subduction	A fault can lock. Strain builds. Sudden slip releases earthquake waves.	Water from a sinking plate helps mantle rock melt. Magma may rise.
Past or transform	Rock can lock then slip and cause earthquakes.	Volcanoes are not usual here.

Japan example

Tohoku region, Japan. 11 March 2011. The Pacific plate sinks beneath northern Japan. Part of the fault locked. Strain built while the plates moved. The fault slipped and released waves, causing a large earthquake. The current record gives magnitude 9.1.

A cause chain

Movement causes a change. That change leads to the hazard. Use because and so to explain the steps. A boundary name alone is not a cause chain.

Pattern and limits

Many hazards lie near plate edges. Some earthquakes occur inside plates. Some volcanoes form at hot spots. Continental collision can cause earthquakes without a chain of volcanoes. A map cannot predict the exact time of the next earthquake.

Where this leads: geoscientists and hazard planners use evidence and maps to explain risk.